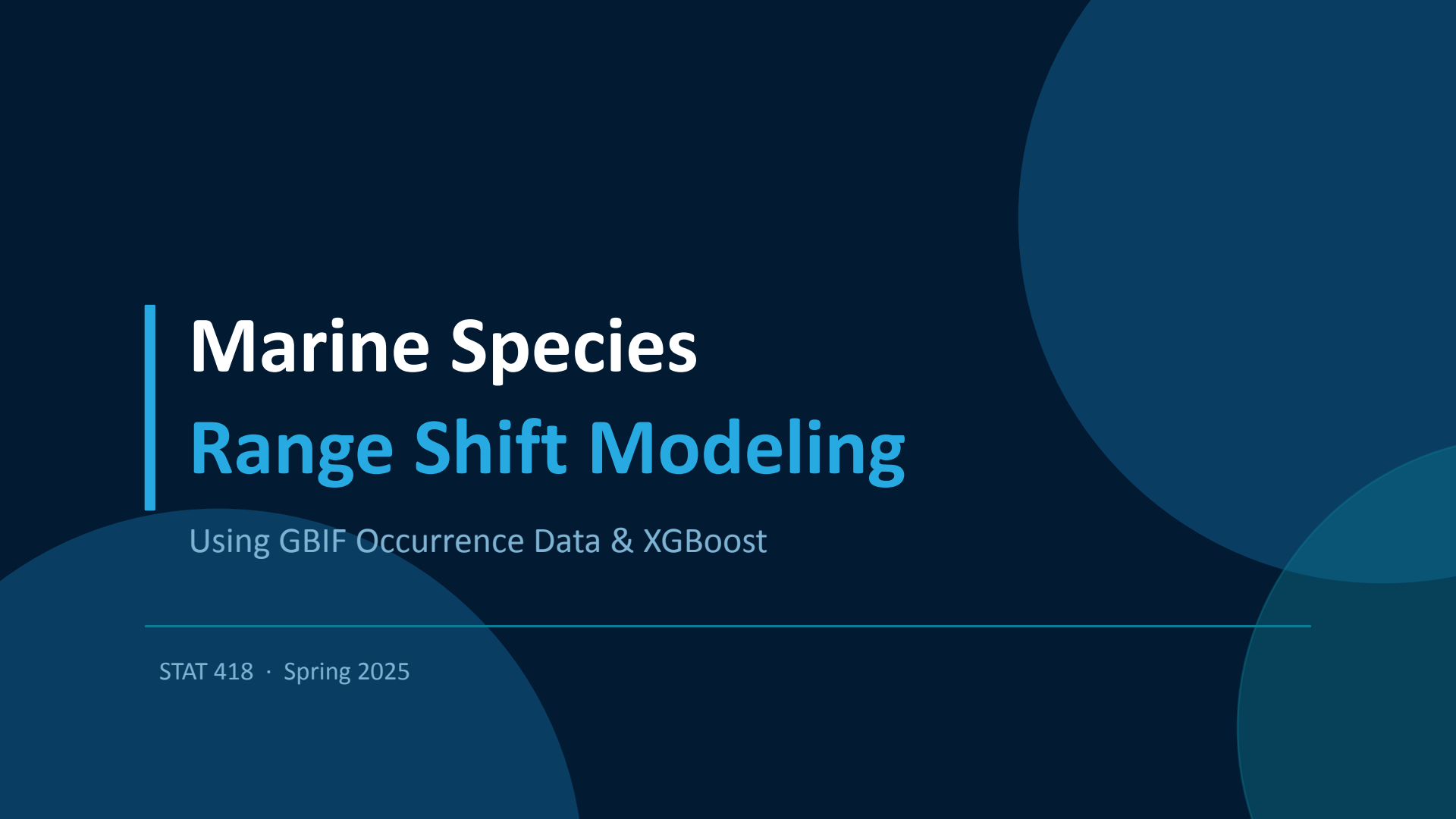




Marine Species Range Shift Modeling

Using GBIF Occurrence Data & XGBoost

STAT 418 · Spring 2025

Project Overview & Motivation

The Problem

Ocean warming is altering where marine species can survive. Traditional monitoring is slow and expensive. Can we use decades of occurrence data to detect and predict range shifts?

Why It Matters

Range shifts affect fisheries, ecosystems, and conservation planning. Early detection enables proactive management before species disappear from historical ranges.

300K+ GBIF Records

45 yrs Historical Coverage

1 Species Northern Gannet

Data Collection Approach

1

GBIF API Data Query

Query “*Morus Bassanus*” (Northern Gannet) from the GBIF API. Split datetime field into year and month.

2

Sampling Strategy

600 records per month $\times$ 12 months $\times$ 45 years = ~324k rows.
Consistent cross-year coverage avoids recency bias from modern citizen science explosion.

3

Bounding Box Filter

North Atlantic only: lat 30–80°N, lon -80–30°E. Removes Pacific vagrant records and southern hemisphere strays.

4

Dataset Quality Filtering

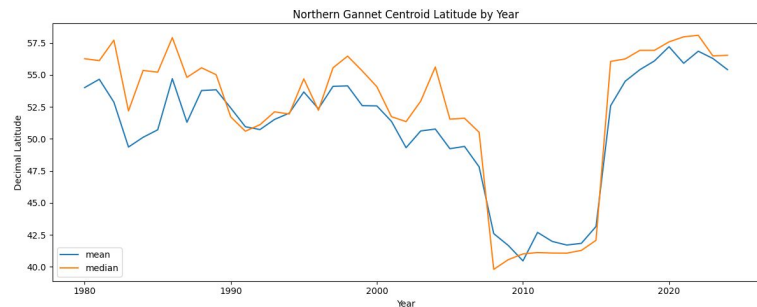
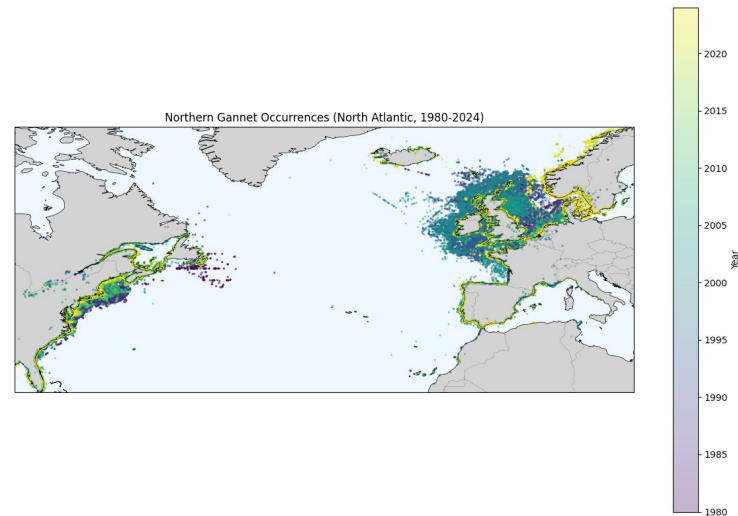
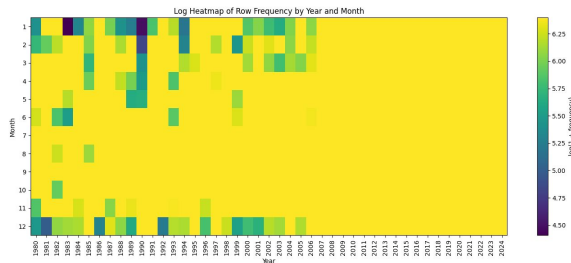
Remove systematic survey grids (PIROP, ICES JCDP, USGS Seabirds DB). Remove miscellaneous data causing biases.

Status: Data collection complete — 324k raw records, cleaning in progress

Exploratory Data Analysis

Key Findings

- Consistent multi-month coverage across all 45 years after month-level sampling
- Core range: UK/Ireland coasts, Bay of Biscay, Canadian Maritimes
- Recent records (post-2010) extend further north into Norwegian Sea & Iceland
- Data problem around 2008-2015, need to filter out problematic rows



Proposed Modeling Approach

Model 1

XGBoost Regression

- Individual records as modeling unit
- Target: decimalLatitude
- Features: year, month_sin/cos, longitude
- Time-based train/test split (pre/post 2015)
- Predict location beyond 2024

Model 2

ConvLSTM

- Spatiotemporal sequence model treating occurrence grids as image-like inputs
- Target: predicted occurrence density grid for next time step (year/season)
- Input: 3D tensor of gridded occurrence counts (lat × lon × time)
- Captures spatial autocorrelation and temporal dependencies simultaneously
- Train on 1980–2015 sequences, evaluate on 2016–2024 held-out period
- Compare predicted vs actual northward density shift over time

Timeline & Next Steps



Data Cleaning

Remove grid datasets, filter out biased rows, finalize 324k clean rows

EDA Complete

Centroid plots, spatial maps, seasonal analysis, leading edge trends

Modeling

Train XGBoost and try ConvLSTM, hyperparameter tuning, SHAP values

API + App

FastAPI model endpoint, Streamlit app with interactive map + predictions

Deploy & Document

Docker, Cloud Run deployment, GitHub README, architecture diagram